

1. Answers to Short Questions

1. **Who created Python and in which year?**

Python was created by Guido van Rossum. It was first released in 1991.

2. **Write the command to check the Python version in Command Prompt.**

The command to check the Python version in Command Prompt is :

```
python --version
```

3. **Define a variable in one line.**

A variable is a name used to store a value in a program.

Example:

```
age = 20
```

4. **Can a Python list store mixed data types?**

Yes. A Python list can store different data types.

Example:

```
data = [10, "Python", 3.5, True]
```

5. **What does int() do?**

The datatype int() converts a value into an integer.

Example:

```
x = int("10")
```

Here, "10" is converted to 10.

6. **Write one example of a Python keyword.**

if is an example of a Python keyword.

Other examples include for, while, def, and class.

7. **What is the index of the first item in a list?**

The index of the first item in a Python list is 0.

Example:

```
numbers = [10, 20, 30]
```

```
# 10 is at index 0
```

8. **Give the syntax of an if statement.**

if condition:

```
    statement
```

9. What does the % operator return?

The % operator returns the remainder after division.

Example:

10 % 3

Output:

1

10. Write a dictionary literal with one key-value pair.

```
student = {"name": "Ram"}
```

Here, name is the key and Ram is its value.

Answers to Long Questions

11. What are the application areas of Python programming?

Python is a popular programming language used in many areas because it is simple and powerful. Its major application areas are:

- Web Development – Used to create websites and web applications using Django and Flask.
- Data Science – Used to analyze and process large amounts of data.
- Artificial Intelligence and Machine Learning – Used to develop AI and machine-learning applications.
- Automation – Used to automate repetitive tasks and processes.
- Desktop Applications – Used to create GUI applications using libraries such as Tkinter.
- Game Development – Used to develop simple games using libraries such as Pygame.
- Scientific Computing – Used for mathematical and scientific calculations.
- Networking and Cybersecurity – Used for network programming, testing, and security tools.

12. Write any two advantages of Python.

The two major advantages of Python are:

- Easy to Learn: Python has simple and readable syntax, making it easy for beginners to learn and use.

- **Large Library Support:** Python provides many built-in and external libraries for web development, data analysis, AI, automation, and other tasks.

13.What is operator precedence? Give a small expression to show it.

Operator precedence is the order in which different operators are evaluated in an expression.

For example:

```
result = 10 + 5 * 2
```

Here, multiplication `*` has higher precedence than addition `+`.

So:

```
10 + (5 × 2)
= 10 + 10
= 20
```

Therefore, the result is 20.

14.What are sequence data types in Python? List any two with examples.

Sequence data types are data types that store items in an ordered sequence. Each item can be accessed using its index.

The two common sequence data types are:

- **List:** A list is an ordered and changeable collection of items.
`numbers = [10, 20, 30]`
- **Tuple:** A tuple is an ordered collection that cannot be changed after creation.
`numbers = (10, 20, 30)`

15.Explain the role of indentation in Python control structures.

Indentation means leaving spaces at the beginning of a line. Python uses indentation to define a block of code.

Example:

```
age = 20
```

```
if age >= 18:
    print("Adult")
```

The indented `print()` statement belongs to the `if` block.

Python generally uses four spaces for indentation. Incorrect indentation can cause an `IndentationError`.

16. Write two rules for naming variables, with one valid and one invalid name.

Two rules for naming variables are:

A variable name must not start with a number.

A variable name can contain letters, numbers, and underscores, but it cannot contain spaces or special symbols.

Valid variable:

```
student_name = "Ram"
```

Invalid variable:

```
2student = "Ram"
```

The second name is invalid because it starts with a number.

17. What is Boolean in Python? Give a simple use case.

Boolean is a data type that has only two possible values:

True

False

It is mainly used for decision-making and conditions.

Example:

```
age = 20
```

```
print(age >= 18)
```

Output:

True

Here, the Boolean result is used to check whether the person is 18 or older.

18. Describe the match-case statement briefly and its version requirement.

The match-case statement is used for matching a value against different patterns. It is similar to the switch statement found in some other programming languages.

Example:

```
day = 1
```

```
match day:
```

```
case 1:
```

```
print("Sunday")
```

```
case 2:
```

```
print("Monday")
```

```
case _:
    print("Invalid day")
```

The match-case statement was introduced in Python 3.10. Therefore, Python 3.10 or later is required to use it.

19.What is the difference between / and //?

The / and // operators are both used for division, but they produce different types of results.

/ – Division operator:

It returns the normal division result, usually as a floating-point number.

```
10 / 3
```

Output:

```
3.3333333333333335
```

// – Floor division operator:

It returns the floor value of the division.

```
10 // 3
```

Output:

```
3
```

20.Explain the flow of a while loop in 2–3 sentences.

A while loop repeatedly executes a block of statements as long as its condition is True. When the condition becomes False, the loop stops.

Example:

```
i = 1
```

```
while i <= 5:
```

```
    print(i)
```

```
    i += 1
```

The loop prints numbers from 1 to 5 and stops when i becomes 6.

21.Define a dictionary and describe how keys and values are written.

A dictionary is a collection of data stored in key-value pairs. Each key is used to access its corresponding value.

Example:

```
student = {
```

```
    "name": "Ram",
```

```
    "age": 20
```

```
}
```

Here:

"name" is a key and "Ram" is its value.

"age" is a key and 20 is its value.

A dictionary is written using curly brackets { }, and keys and values are separated by a colon :.

3. Debug the Following Programs:

Debug 1 — Corrected

```
print("Hello Python")
name = input("Name?")
if len(name) > 0:
    print("Hi, ", name)
```

Debug 2 — Corrected

```
a = int(input("A: "))
b = int(input("B: "))
sum = a + b
```

```
print("Sum =", sum)
print("Done")
```

```
if a > b:
    print("A bigger")
else:
    print("B bigger")
```

Debug 3 — Corrected

```
nums = [1, 2, 3]

for n in nums:
    print(n)
    n += 1

print(n)

match n:
    case 3:
        print("three")
```

Debug 4 — Corrected

```
x = 10
y = 0

if y == 0:
    print("Cannot divide by zero")
else:
    print(x / y)

print("done")
```

Debug 5 — Corrected

```
data = {"Nepal": "Rupee", "USA": "Dollar"}

print(data["Nepal"])

data["USA"] = "USD"
data["Nepal"] = "NPR"

if "USA" in data:
    print("ok")
```

Debug 6 — Corrected

```
a, b, c = 5, 10, 15
```

```
if a > b and b > c:  
    print("a")  
elif b > c:  
    print("b")  
else:  
    print("c")
```

Debug 7 — Corrected

```
for i in range(3):  
    print(i)  
    i += 1
```

```
while i < 3:  
    print("loop")  
    i += 1
```

Debug 8 — Corrected

```
text = "Python"
```

```
if text.lower() == "python":  
    print("ok")  
elif text == "Py":  
    print("short")  
else:  
    print("no")
```

```
match text:  
    case "Python":  
        print("Title")  
    case _:  
        print("other")
```

```
print("done")
```

Debug 9 — Corrected

```
a = int(input("A: "))  
b = int(input("B: "))
```

```
c = a + b  
print("Sum =", c)
```

```
if c > 10:  
    print("big")  
else:  
    print("small")
```

Debug 10 — Corrected

```
lst = [1, 2, 3]
```

```
print(lst[2])  
lst.append(4)
```

```
for n in lst:  
    print(n)
```

```
if len(lst) >= 3:  
    print("ok")
```

Debug 11 — Corrected

```
def add(a, b):  
    return a + b
```

```
print(add(2, 3))
```

```
def show(msg):  
    print(msg)
```

```
show(100 + 2)
```

Debug 12 — Corrected

```
day = 2

match day:
    case 1:
        print("Sun")
    case 2:
        print("Mon")
    case 3:
        print("Tue")
    case 4:
        print("Wed")
    case 5:
        print("Wed")
    case _:
        print("Other")
```

4. Write a Program Using Python To:

A. Find the area of a rectangular box.

```
length = float(input("Enter length: "))
breadth = float(input("Enter breadth: "))
area = length * breadth

print("Area =", area)
```

B. Read the purchase amount and print a 5% discount if the amount > 20000.

```
amount = float(input("Enter purchase amount: "))
if amount > 20000:
    discount = amount * 5 / 100
    print("Discount =", discount)
else:
    print("No discount")
```

C. Check if an input number is positive, negative, or zero.

```
num = int(input("Enter a number: "))
if num > 0:
    print("Positive")
elif num < 0:
    print("Negative")
else:
    print("Zero")
```

D. y. Input three numbers and print the smallest number.

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))
c = int(input("Enter third number: "))
if a < b and a < c:
    print("Smallest =", a)
elif b < c:
    print("Smallest =", b)
else:
    print("Smallest =", c)
```

E. z. Print the name of the month based on the number.

```
month = int(input("Enter month number: "))
match month:
    case 1:
        print("January")
    case 2:
        print("February")
    case 3:
        print("March")
    case 4:
        print("April")
    case 5:
        print("May")
    case 6:
        print("June")
    case 7:
        print("July")
```

```
case 8:
    print("August")
case 9:
    print("September")
case 10:
    print("October")
case 11:
    print("November")
case 12:
    print("December")
case _:
    print("Invalid month")
```

F. Read a list of five integers and print the item at index 2.

```
lst = []

for i in range(5):
    num = int(input("Enter number: "))
    lst.append(num)

print("Item at index 2 =", lst[2])
```

G. Create a dictionary of three countries and currencies, then print the currency of "Nepal".

```
data = {
    "Nepal": "NPR",
    "India": "INR",
    "USA": "USD"
}

print("Currency of Nepal =", data["Nepal"])
```

H. Using a for loop, print numbers from 1 to 10.

```
for i in range(1, 11):  
    print(i)
```

I. Using a while loop, print the sum of integers from 1 to n.

```
n = int(input("Enter n: "))  
i = 1  
sum = 0  
while i <= n:  
    sum = sum + i  
    i += 1  
print("Sum =", sum)
```